MAGNETISM AND MATTER

CHAPTER FIVE
CLASS - XII

MANISH JOSHI

D.A.V. LOHAGHAT

July 23, 2025

Learning Objectives:

- Bar magnet, Magnetic field lines

Learning Objectives:

- Bar magnet, Magnetic field lines
- Bar magnet as an equivalent solenoid (qualitative treatment only).

Learning Objectives:

- Bar magnet, Magnetic field lines
- Bar magnet as an equivalent solenoid (qualitative treatment only).
- Torque on a magnetic dipole (bar magnet) in a uniform magnetic field (qualitative treatment only).

Learning Objectives:

- Bar magnet, Magnetic field lines
- Bar magnet as an equivalent solenoid (qualitative treatment only).
- Torque on a magnetic dipole (bar magnet) in a uniform magnetic field (qualitative treatment only).
- Magnetic field intensity due to a magnetic dipole (bar magnet) along its axis and perpendicular to its axis (qualitative treatment only).

Learning Objectives:

- Bar magnet, Magnetic field lines
- Bar magnet as an equivalent solenoid (qualitative treatment only).
- Torque on a magnetic dipole (bar magnet) in a uniform magnetic field (qualitative treatment only).
- Magnetic field intensity due to a magnetic dipole (bar magnet) along its axis and perpendicular to its axis (qualitative treatment only).
- Magnetization of materials.

Learning Objectives:

- Bar magnet, Magnetic field lines
- Bar magnet as an equivalent solenoid (qualitative treatment only).
- Torque on a magnetic dipole (bar magnet) in a uniform magnetic field (qualitative treatment only).
- Magnetic field intensity due to a magnetic dipole (bar magnet) along its axis and perpendicular to its axis (qualitative treatment only).
- Magnetization of materials.
- Magnetic properties of materials- Para-, dia- and ferro-magnetic substances with examples.

Learning Objectives:

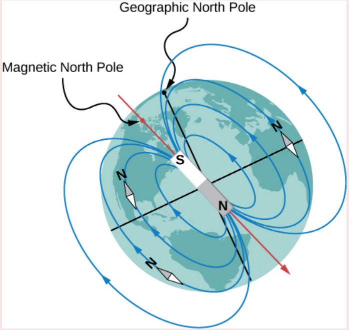
- Bar magnet, Magnetic field lines
- Bar magnet as an equivalent solenoid (qualitative treatment only).
- Torque on a magnetic dipole (bar magnet) in a uniform magnetic field (qualitative treatment only).
- Magnetic field intensity due to a magnetic dipole (bar magnet) along its axis and perpendicular to its axis (qualitative treatment only).
- Magnetization of materials.
- Magnetic properties of materials- Para-, dia- and ferro-magnetic substances with examples.
- Effect of temperature on magnetic properties.

Learning Objectives:

- Bar magnet, Magnetic field lines
- Bar magnet as an equivalent solenoid (qualitative treatment only).
- Torque on a magnetic dipole (bar magnet) in a uniform magnetic field (qualitative treatment only).
- Magnetic field intensity due to a magnetic dipole (bar magnet) along its axis and perpendicular to its axis (qualitative treatment only).
- Magnetization of materials.
- Magnetic properties of materials- Para-, dia- and ferro-magnetic substances with examples.
- Effect of temperature on magnetic properties.

Introduction

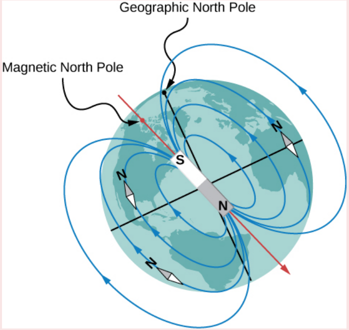
Some of the commonly known ideas regarding magnetism are:



- The **earth behaves as a magnet** with the magnetic field pointing approximately from the geographic south to the north.

Introduction

Some of the commonly known ideas regarding magnetism are:



- The **earth behaves as a magnet** with the magnetic field pointing approximately from the geographic south to the north.
- When a **bar magnet** is **freely suspended**, it **points in the north-south direction**. The tip which points to the **geographic north** is called the **north pole** and the tip which points to the **geographic south** is called the **south pole** of the magnet.

Introduction

Some of the commonly known ideas regarding magnetism are:

- Magnetic poles **repel** if they are **alike** (both N or both S), they **attract** if they are **opposite** (one N and the other S)

Introduction

Some of the commonly known ideas regarding magnetism are:

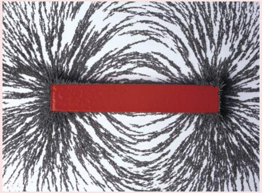
- Magnetic poles **repel** if they are **alike** (both N or both S), they **attract** if they are **opposite** (one N and the other S)
- **We cannot isolate the north, or south pole** of a magnet. If a bar magnet is broken into two halves, we get two similar bar magnets with somewhat weaker properties. **Unlike electric charges**, isolated magnetic north and south poles known as **magnetic monopoles do not exist**.

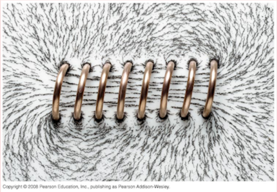
Introduction

Some of the commonly known ideas regarding magnetism are:

- Magnetic poles **repel** if they are **alike** (both N or both S), they **attract** if they are **opposite** (one N and the other S)
- **We cannot isolate the north, or south pole** of a magnet. If a bar magnet is broken into two halves, we get two similar bar magnets with somewhat weaker properties. **Unlike electric charges**, isolated magnetic north and south poles known as **magnetic monopoles do not exist**.
- It is **possible** to **make magnets** out of **iron and its alloys**.

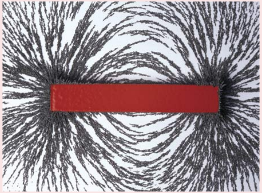
THE BAR MAGNET AND MAGNETIC FIELD LINES

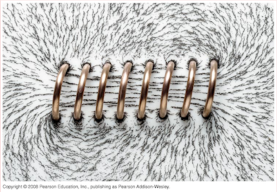




- By examining the **pattern of iron filings** sprinkled on a sheet of glass placed over a short **bar magnet**.

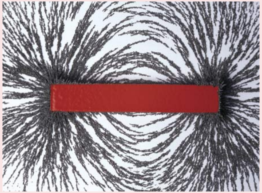
THE BAR MAGNET AND MAGNETIC FIELD LINES

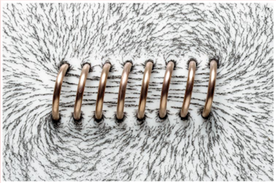




- By examining the **pattern of iron filings** sprinkled on a sheet of glass placed over a short **bar magnet**.
- The **pattern** of iron filings **suggests** that the **magnet has two poles. similar to** the **positive and negative charge** of an electric dipole.

THE BAR MAGNET AND MAGNETIC FIELD LINES





Copyright © 2008 Pearson Education, Inc., publishing as Pearson Addison-Wesley.

- By examining the **pattern of iron filings** sprinkled on a sheet of glass placed over a short **bar magnet**.
- The **pattern** of iron filings **suggests** that the **magnet has two poles. similar to** the **positive and negative charge** of an electric dipole.
- A **similar pattern** of iron filings is **observed** around a current carrying **solenoid**.

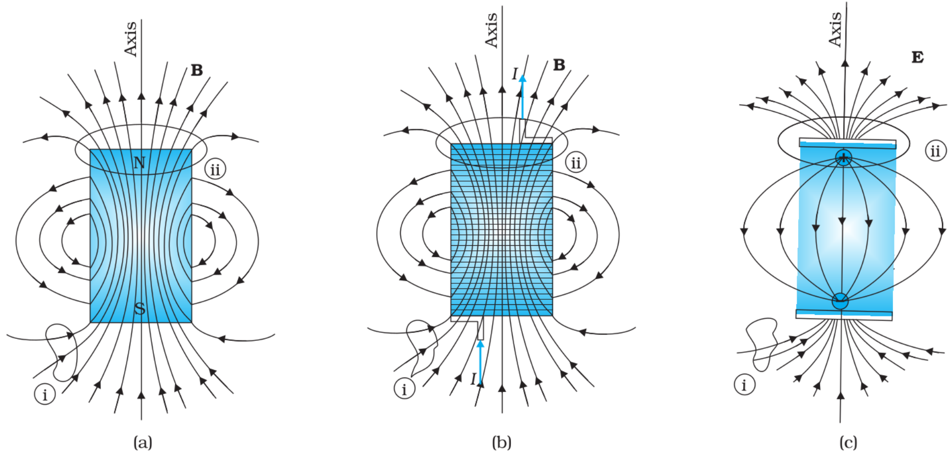


FIGURE 5.3 The field lines of (a) a bar magnet, (b) a current-carrying finite solenoid and (c) electric dipole. At large distances, the field lines are very similar. The curves labelled (i) and (ii) are closed Gaussian surfaces.

THE BAR MAGNET AND MAGNETIC FIELD LINES

The **magnetic field lines are** a **visual and intuitive realisation of** the **magnetic field**. Their **properties** are:

THE BAR MAGNET AND MAGNETIC FIELD LINES

The **magnetic field lines are** a **visual and intuitive realisation of** the **magnetic field**. Their **properties** are:

- The **magnetic field lines** of a magnet (or a solenoid) **form continuous closed loops**. This is **unlike the electric dipole** where these field lines **begin from a positive charge** and **end on the negative charge or escape to infinity**.

THE BAR MAGNET AND MAGNETIC FIELD LINES

The **magnetic field lines are** a **visual and intuitive realisation of** the **magnetic field**. Their **properties** are:

- The **magnetic field lines** of a magnet (or a solenoid) **form continuous closed loops**. This is **unlike the electric dipole** where these field lines **begin from a positive charge** and **end on the negative charge or escape to infinity**.
- The **tangent to the field line** at a given point **represents the direction of** the **net magnetic field B** at that point.

THE BAR MAGNET AND MAGNETIC FIELD LINES

The **magnetic field lines are** a **visual and intuitive realisation of** the **magnetic field**. Their **properties** are:

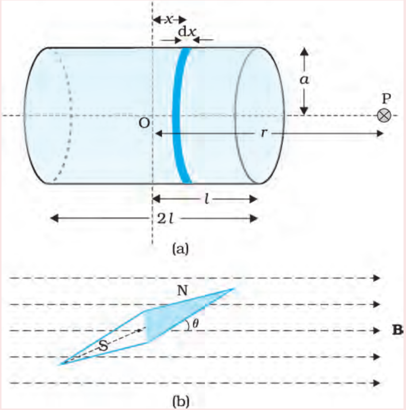
- The **magnetic field lines** of a magnet (or a solenoid) **form continuous closed loops**. This is **unlike the electric dipole** where these field lines **begin from a positive charge** and **end on the negative charge or escape to infinity**.
- The **tangent to the field line** at a given point **represents the direction of** the **net magnetic field B** at that point.
- The **larger the number of field lines crossing per unit area**, the **stronger is the magnitude of the magnetic field B**.

THE BAR MAGNET AND MAGNETIC FIELD LINES

The **magnetic field lines are** a **visual and intuitive realisation of** the **magnetic field**. Their **properties** are:

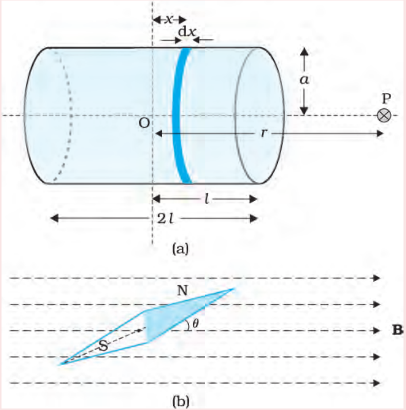
- The **magnetic field lines** of a magnet (or a solenoid) **form continuous closed loops**. This is **unlike the electric dipole** where these field lines **begin from a positive charge** and **end on the negative charge or escape to infinity**.
- The **tangent to the field line** at a given point **represents the direction of** the **net magnetic field B** at that point.
- The **larger the number of field lines crossing per unit area**, the **stronger is the magnitude of the magnetic field B**.
- The **magnetic field lines do not intersect**, for **if they did**, the **direction of the magnetic field would not be unique** at the point of intersection.

Bar magnet as an equivalent solenoid



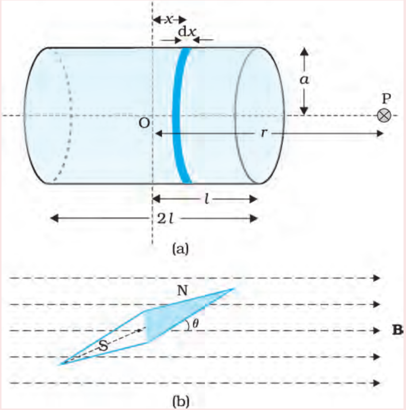
- The **resemblance of magnetic field lines** for a **bar magnet** and a **solenoid suggest** that a **bar magnet may be thought of** as a **large number of circulating currents** in **analogy** with a **solenoid**.

Bar magnet as an equivalent solenoid



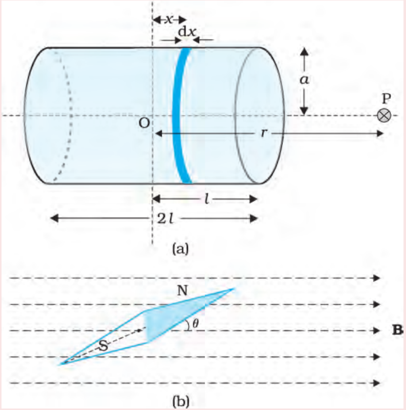
- The **resemblance of magnetic field lines** for a **bar magnet** and a **solenoid suggest** that a **bar magnet may be thought of** as a **large number of circulating currents** in **analogy** with a **solenoid**.
- To make this analogy more firm **we calculate the axial field** of a **finite solenoid**.

Bar magnet as an equivalent solenoid



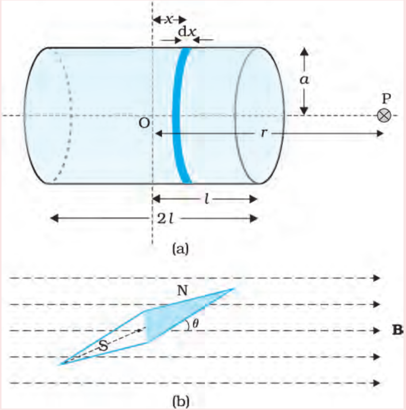
- The **resemblance of magnetic field lines** for a **bar magnet** and a **solenoid suggest** that a **bar magnet may be thought of** as a **large number of circulating currents** in **analogy** with a **solenoid**.
- To make this analogy more firm **we calculate the axial field** of a **finite solenoid**.
- We shall **demonstrate** that **at large distances** this **axial field resembles** that of a **bar magnet**.

Bar magnet as an equivalent solenoid



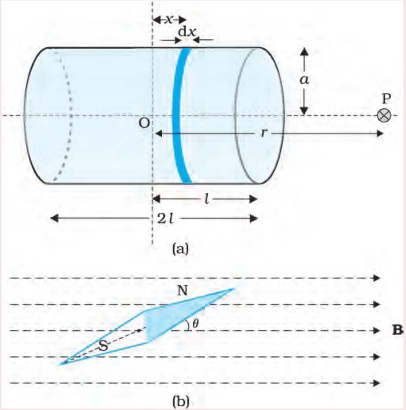
- Let, n be the **number of turns per unit length**.

Bar magnet as an equivalent solenoid



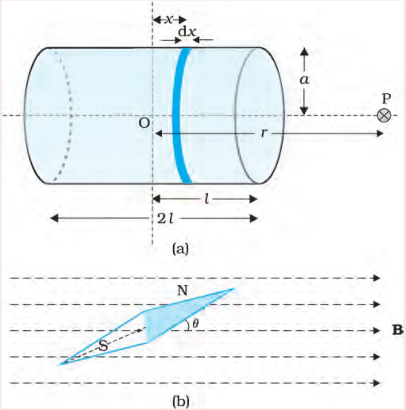
- Let, n be the **number of turns per unit length**.
- Let solenoid **length** be $\mathbf{2}l$ and **radius** a.

Bar magnet as an equivalent solenoid



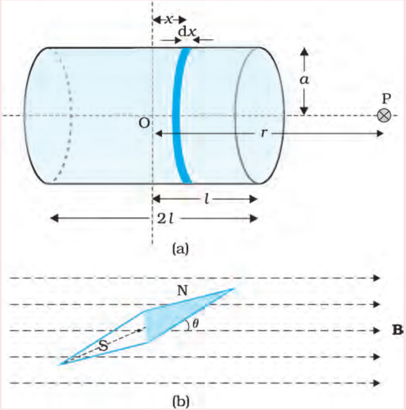
- Let, n be the **number of turns per unit length**.
- Let solenoid **length** be $\mathbf{2}l$ and **radius** a.
- We want to **evaluate** the **axial field at a point P**, at a **distance r from** the **centre O** of the solenoid.

Bar magnet as an equivalent solenoid



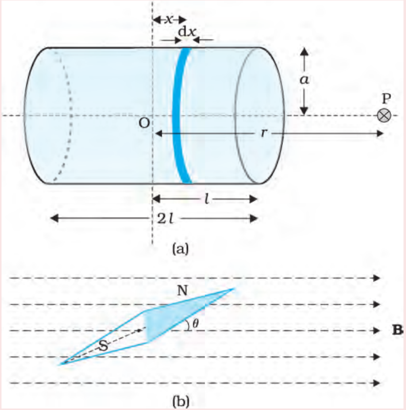
- Let, n be the **number of turns per unit length**.
- Let solenoid **length** be $2l$ and **radius** a.
- We want to **evaluate** the **axial field at a point P**, at a **distance r from** the **centre O** of the solenoid.
- **consider a circular element** of **thickness** dx of the solenoid **at a distance** x from its centre. It **consists** of ndx turns. Let I be the **current** in the solenoid.

Bar magnet as an equivalent solenoid



- The **magnitude of the field** at **point P due to** the **circular element** is

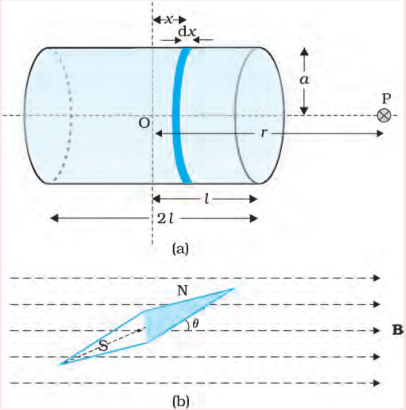
Bar magnet as an equivalent solenoid



- The **magnitude of the field** at **point P due to** the **circular element** is

$$dB = \frac{\mu_0 I a^2 n \, dx}{2\left[(r-x)^2 + a^2\right]^{3/2}}$$

Bar magnet as an equivalent solenoid

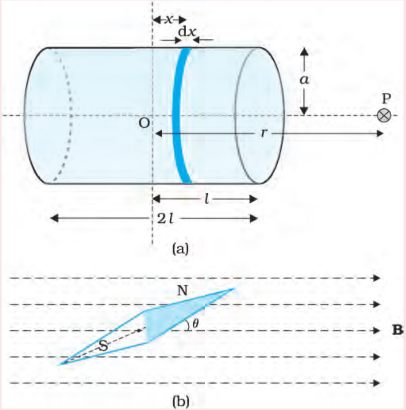


- The **magnitude of the field** at **point P due to** the **circular element** is

$$dB = \frac{\mu_0 I a^2 n\, dx}{2\left[(r-x)^2 + a^2\right]^{3/2}}$$

- The magnitude of the total field is obtained by integrating from $x = -l$ to $x = +l$. Thus,

Bar magnet as an equivalent solenoid



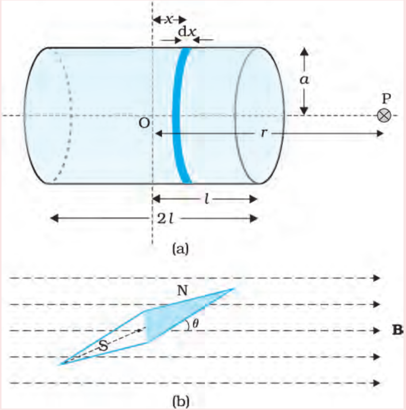
- The **magnitude of the field** at **point P due to** the **circular element** is

$$\boldsymbol{dB = \frac{\mu_0 I a^2 n\, dx}{2\left[(r-x)^2 + a^2\right]^{3/2}}}$$

- The magnitude of the total field is obtained by integrating from $\boldsymbol{x = -l}$ to $\boldsymbol{x = +l}$. Thus,

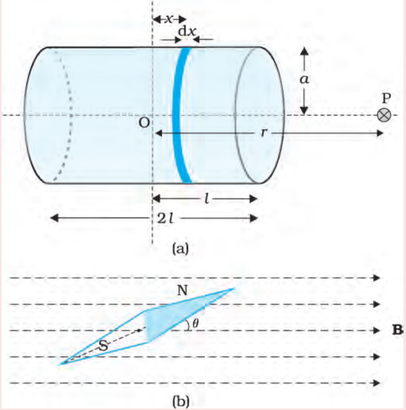
$$B = \frac{\mu_0 I a^2 n}{2} \int_{-l}^{l} \frac{dx}{\left[(r-x)^2 + a^2\right]^{3/2}}$$

Bar magnet as an equivalent solenoid



- This integration can be done by trigonometric substitution. But for far axial points **i.e.,** for $r >> x$, and $r >> a$ this expression simplifies to

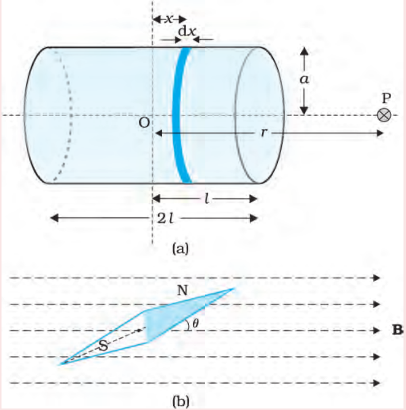
Bar magnet as an equivalent solenoid



- This integration can be done by trigonometric substitution. But for far axial points **i.e.,** for $\boldsymbol{r >> x}$, and $\boldsymbol{r >> a}$ this expression simplifies to

$$B = \frac{\mu_0 I a^2 n}{2} \int_{-l}^{l} \frac{dx}{[r^2]^{3/2}}$$

Bar magnet as an equivalent solenoid

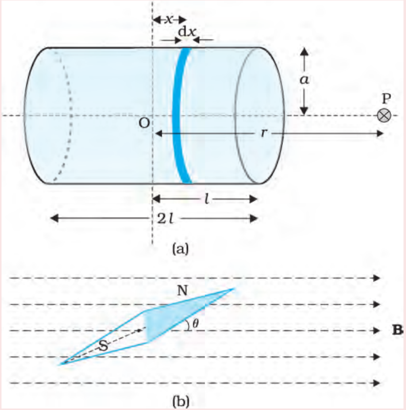


- This integration can be done by trigonometric substitution. But for far axial points **i.e.,** for $\boldsymbol{r >> x}$, and $\boldsymbol{r >> a}$ this expression simplifies to

$$B = \frac{\mu_0 I a^2 n}{2} \int_{-l}^{l} \frac{dx}{[r^2]^{3/2}}$$

$$B = \frac{\mu_0 I a^2 n}{2r^3} \int_{-l}^{l} dx = \frac{\mu_0 I a^2 n}{2r^3} \times 2l$$

Bar magnet as an equivalent solenoid



- This integration can be done by trigonometric substitution. But for far axial points **i.e.,** for $\boldsymbol{r >> x}$, and $\boldsymbol{r >> a}$ this expression simplifies to

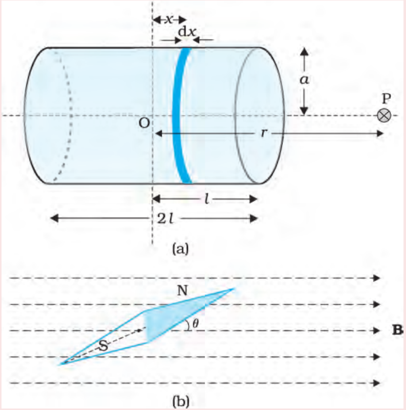
$$B = \frac{\mu_0 I a^2 n}{2} \int_{-l}^{l} \frac{dx}{[r^2]^{3/2}}$$

$$B = \frac{\mu_0 I a^2 n}{2r^3} \int_{-l}^{l} dx = \frac{\mu_0 I a^2 n}{2r^3} \times 2l$$

- magnetic moment of solenoid

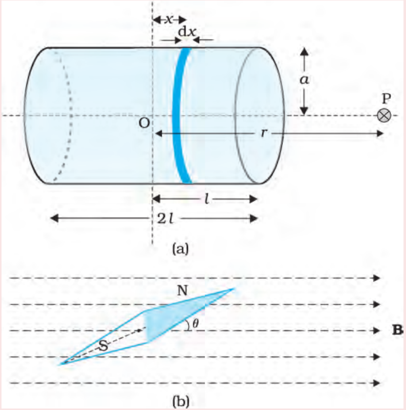
$$\boxed{\boldsymbol{m = (2nlI)(\pi a^2)}}$$

Bar magnet as an equivalent solenoid



$$\boxed{B = \frac{\mu_0}{4\pi}\frac{2m}{r^3}}$$

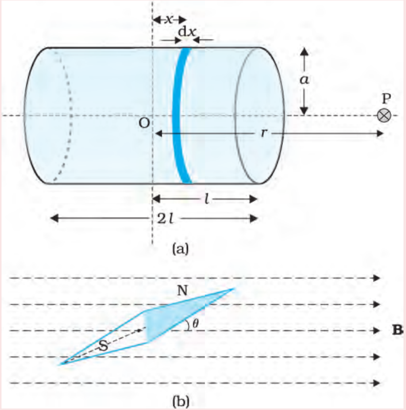
Bar magnet as an equivalent solenoid



$$\boxed{B = \frac{\mu_0}{4\pi} \frac{2m}{r^3}}$$

- **This is also** the **far axial magnetic field of a bar magnet** which one may obtain experimentally.

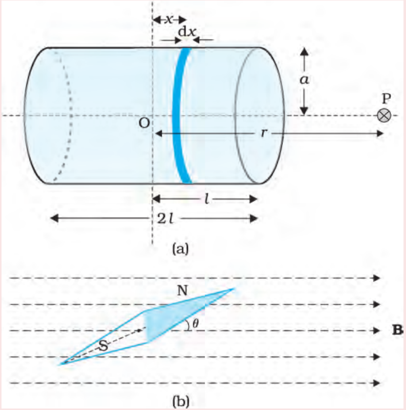
Bar magnet as an equivalent solenoid



$$\boxed{B = \frac{\mu_0}{4\pi}\frac{2m}{r^3}}$$

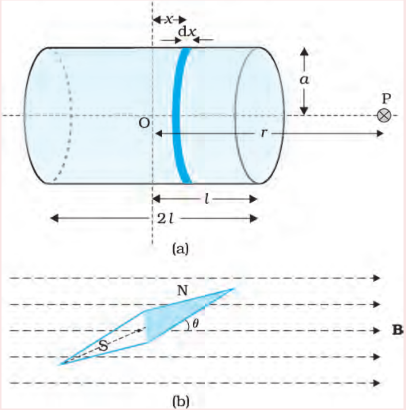
- **This is also** the **far axial magnetic field of a bar magnet** which one may obtain experimentally.
- Thus, a **bar magnet and a solenoid produce similar magnetic field.**

The dipole in a uniform magnetic field



- When we place a magnetic dipole (magnetic needle) of known magnetic moment $\vec{m}$ in the magnetic filed $\vec{B}$.

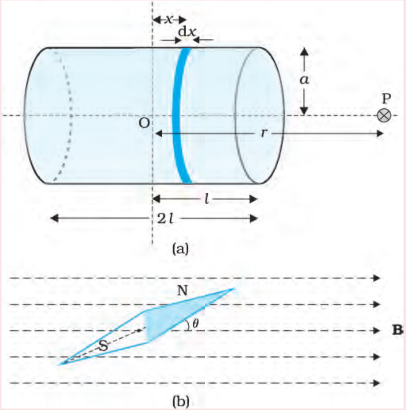
The dipole in a uniform magnetic field



- When we place a magnetic dipole (magnetic needle) of known magnetic moment $\vec{m}$ in the magnetic filed $\vec{B}$.
- The **torque in the magnetic dipole** (needle) is

$$\boxed{\vec{\tau} = \vec{m} \times \vec{B}}$$

The dipole in a uniform magnetic field

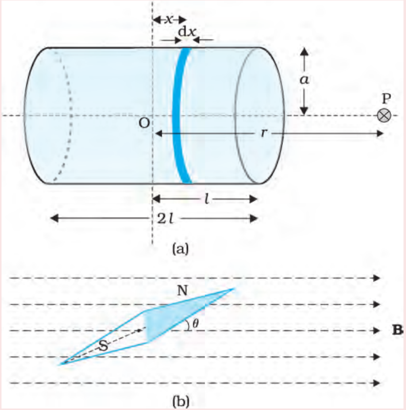


- When we place a magnetic dipole (magnetic needle) of known magnetic moment $\vec{m}$ in the magnetic filed $\vec{B}$.
- The **torque in the magnetic dipole** (needle) is

$$\boxed{\vec{\tau} = \vec{m} \times \vec{B}}$$

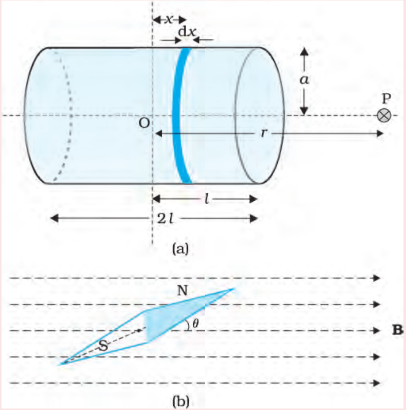
- In magnitude $\tau = mB\sin\theta$

The dipole in a uniform magnetic field



- An **expression for magnetic potential energy** U_m can also be **obtained** on lines **similar to electrostatic potential energy**.

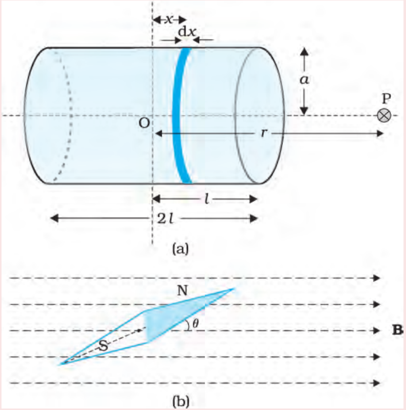
The dipole in a uniform magnetic field



- An **expression for magnetic potential energy** U_m can also be **obtained** on lines **similar to electrostatic potential energy**.

$$U_m = \int \tau(\theta)\, d\theta$$

The dipole in a uniform magnetic field

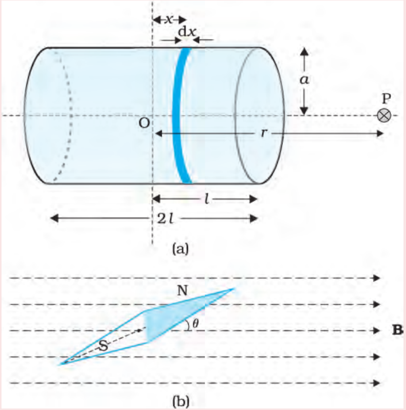


- An **expression for magnetic potential energy** U_m can also be **obtained** on lines **similar to electrostatic potential energy**.

$$U_m = \int \tau(\theta)\, d\theta$$

$$U_m = \int mB\ \sin\theta\ d\theta = -mB\ \cos\theta$$

The dipole in a uniform magnetic field



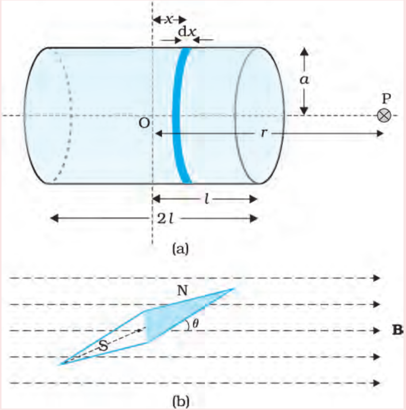
- An **expression for magnetic potential energy** U_m can also be **obtained** on lines **similar to electrostatic potential energy**.

$$U_m = \int \tau(\theta)\, d\theta$$

$$U_m = \int mB\ \sin\theta\, d\theta = -mB\ \cos\theta$$

$$\boxed{U_m = -\vec{m}\cdot\vec{B}}$$

The dipole in a uniform magnetic field



- An **expression for magnetic potential energy** U_m can also be **obtained** on lines **similar to electrostatic potential energy**.

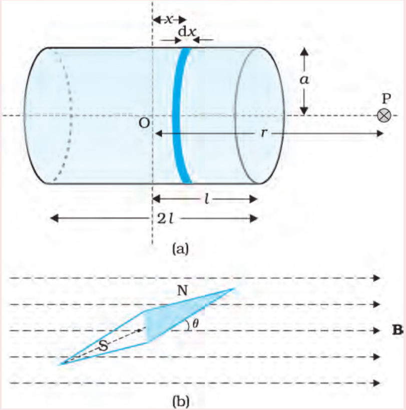
$$U_m = \int \tau(\theta)\, d\theta$$

$$U_m = \int mB\ \sin\theta\ d\theta = -mB\ \cos\theta$$

$$\boxed{U_m = -\vec{m}\cdot\vec{B}}$$

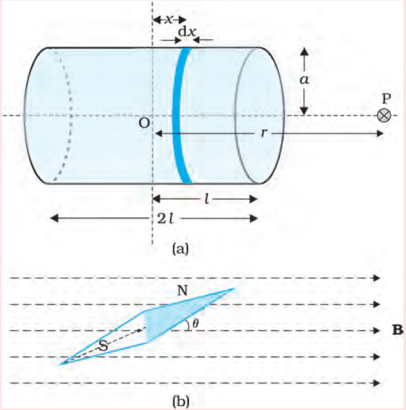
- We can consider **zero potential energy** at $\theta = 90°$

The dipole in a uniform magnetic field



- **potential energy is minimum** ($U_{min} = -mB$) at $\theta = 0°$ (**most stable position**) and

The dipole in a uniform magnetic field



- **potential energy is minimum** $(U_{min} = -mB)$ at $\theta = 0°$ (**most stable position**) and
- **maximum** $(U_{max} = +mB)$ at $\theta = 180°$ (**most unstable position**).

The analogy between electric and magnetic dipoles.

	Electrostatics	Magnetism
	$\dfrac{1}{\epsilon_0}$	μ_0
Dipole moment	$\vec{p}$	$\vec{m}$
Equatorial Field for a short dipole	$\vec{E}_{eq} = \dfrac{1}{4\pi\epsilon_0}\dfrac{\vec{p}}{r^3}$	$\vec{B}_{eq} = \dfrac{\mu_0}{4\pi}\dfrac{\vec{m}}{r^3}$
Axial Field for a short dipole	$\vec{E}_{axial} = \dfrac{1}{4\pi\epsilon_0}\dfrac{2\vec{p}}{r^3}$	$\vec{B}_{axial} = \dfrac{1}{4\pi\epsilon_0}\dfrac{2\vec{m}}{r^3}$
External Field: Torque	$\vec{\tau} = \vec{p} \times \vec{E}$	$\vec{\tau} = \vec{m} \times \vec{B}$
External Field: Energy	$U = -\vec{p} \cdot \vec{E}$	$U = -\vec{m} \cdot \vec{B}$

Note: Do all NCERT exercise questions related to this topic in your notebook

5.3 A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.25 T experiences a torque of magnitude equal to 4.5×10^{-2} J. What is the magnitude of magnetic moment of the magnet?

5.4 A short bar magnet of magnetic moment $m = 0.32\ JT^{-1}$ is placed in a uniform magnetic field of 0.15 T. If the bar is free to rotate in the plane of the field, which orientation would correspond to its (a) stable, and (b) unstable equilibrium? What is the potential energy of the magnet in each case?

5.5 A closely wound solenoid of 800 turns and area of cross section 2.5×10^{-4} m^2 carries a current of 3.0 A. Explain the sense in which the solenoid acts like a bar magnet. What is its associated magnetic moment?

5.6 If the solenoid in Exercise 5.5 is free to turn about the vertical direction and a uniform horizontal magnetic field of 0.25 T is applied, what is the magnitude of torque on the solenoid when its axis makes an angle of 30° with the direction of applied field?

5.7 A bar magnet of magnetic moment 1.5 J T^{-1} lies aligned with the direction of a uniform magnetic field of 0.22 T.

(a) What is the amount of work required by an external torque to turn the magnet so as to align its magnetic moment: (i) normal to the field direction, (ii) opposite to the field direction?

(b) What is the torque on the magnet in cases (i) and (ii)?

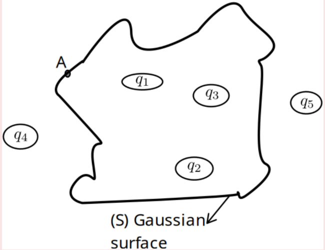
5.8 A closely wound solenoid of 2000 turns and area of cross-section $1.6 \times 10^{-4}\ m^2$, carrying a current of 4.0 A, is suspended through its centre allowing it to turn in a horizontal plane.

(a) What is the magnetic moment associated with the solenoid?

(b) What is the force and torque on the solenoid if a uniform horizontal magnetic field of 7.5×10^{-2} T is set up at an angle of 30° with the axis of the solenoid?

5.12 A short bar magnet has a magnetic moment of 0.48 J T^{-1}. Give the direction and magnitude of the magnetic field produced by the magnet at a distance of 10 cm from the centre of the magnet on (a) the axis, (b) the equatorial lines (normal bisector) of the magnet.

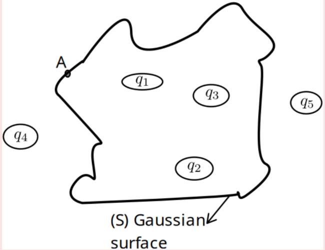
MAGNETISM AND GAUSS'S LAW



- **Gauss's law for magnetism** is: The **net magnetic flux through any closed surface is zero**.

$$\oint \vec{B} \cdot d\vec{S} = 0$$

MAGNETISM AND GAUSS'S LAW

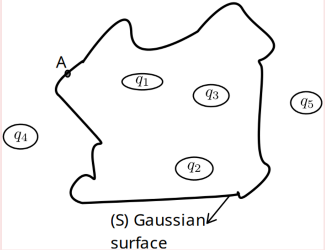


- **Gauss's law for magnetism** is: The **net magnetic flux through any closed surface is zero**.

$$\oint \vec{B} \cdot d\vec{S} = 0$$

- Number of **magnetic field lines entering** the closed surface is **equal** to the **number of field lines leaving** the surface.

MAGNETISM AND GAUSS'S LAW

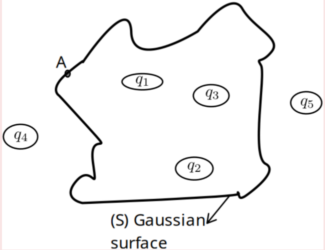


- **Gauss's law for magnetism** is: The **net magnetic flux through any closed surface is zero**.

$$\oint \vec{B} \cdot d\vec{S} = 0$$

- Number of **magnetic field lines entering** the closed surface is **equal** to the **number of field lines leaving** the surface.
- This law is consistent with the observation that **isolated magnetic poles (monopoles) do not exist**.

MAGNETISM AND GAUSS'S LAW



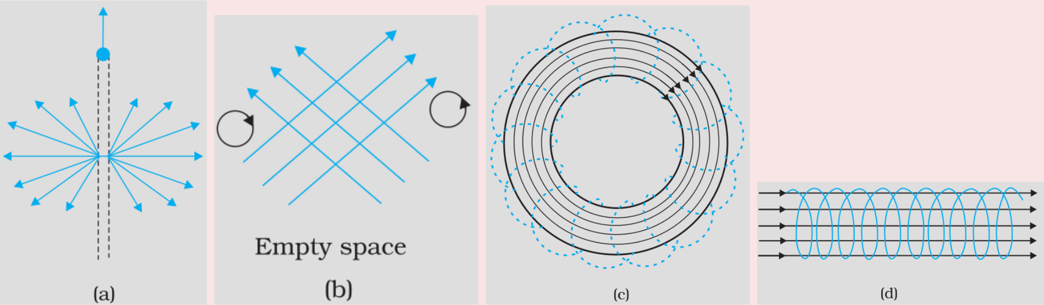
- **Gauss's law for magnetism** is: The **net magnetic flux through any closed surface is zero**.

$$\oint \vec{B} \cdot d\vec{S} = 0$$

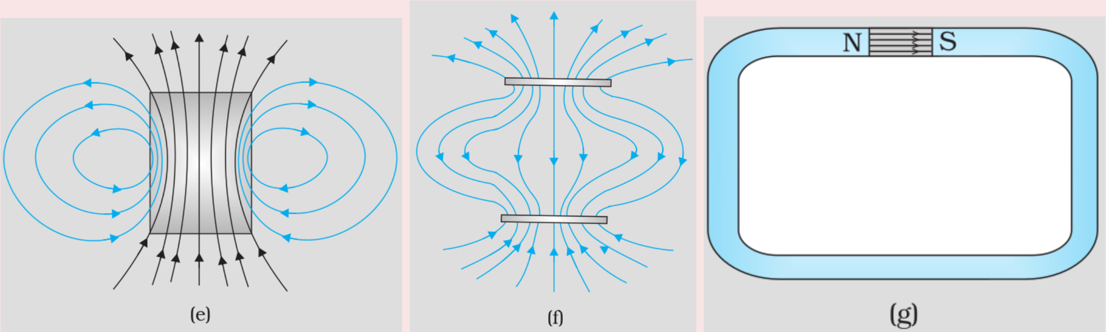
- Number of **magnetic field lines entering** the closed surface is **equal** to the **number of field lines leaving** the surface.
- This law is consistent with the observation that **isolated magnetic poles (monopoles) do not exist**.
- There are **no sources or sinks of magnetic field.**

Example 5.6

Many of the diagrams given in Fig. 5.7 show magnetic field lines (thick lines in the figure) wrongly. Point out what is wrong with them. Some of them may describe electrostatic field lines correctly. Point out which ones.



Example 5.6



(e) (f)

Example 5.7

1. Magnetic field lines show the direction (at every point) along which a small magnetised needle aligns (at the point). Do the magnetic field lines also represent the lines of force on a moving charged particle at every point?
2. Magnetic field lines can be entirely confined within the core of a toroid, but not within a straight solenoid. Why?
3. If magnetic monopoles existed, how would the Gauss's law of magnetism be modified?
4. Does a bar magnet exert a torque on itself due to its own field? Does one element of a current-carrying wire exert a force on another element of the same wire?
5. Magnetic field arises due to charges in motion. Can a system have magnetic moments even though its net charge is zero?